

torch.normal#

<https://pytorch.org/docs/stable/generated/torch.normal.html#torch.normal>

Description:

Returns a tensor of random numbers drawn from separate normal distributions whose mean and standard deviation are given. The mean is a tensor with the mean of each output element's normal distribution The std is a tensor with the standard deviation of each output element's normal distribution The shapes of mean and std don't need to match, but the total number of elements in each tensor need to be the same.

Function Signature

```
torch.normal(mean, std, *, generator=None, out=None) → Tensor#
```

Parameters

Keyword Arguments

```
torch.normal(mean=0.0, std, *, out=None) → Tensor
```

Parameters

Parameters

Keyword

generator (torch.Generator, optional) – a pseudorandom number generator for sampling out (Tensor, optional) – the output tensor.

Keyword

out (Tensor, optional) – the output tensor.

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out (Tensor, optional) – the output tensor.

Code Examples

```
>>> torch.normal(mean=torch.arange(1., 11.), std=torch.arange(1, 0, -0.1))
tensor([ 1.0425,  3.5672,  2.7969,  4.2925,  4.7229,
        6.2134,
         8.0505,  8.1408,  9.0563, 10.0566])
```

```
>>> torch.normal(mean=0.5, std=torch.arange(1., 6.))
tensor([-1.2793, -1.0732, -2.0687,  5.1177, -1.2303])
```

```
>>> torch.normal(mean=torch.arange(1., 6.))
tensor([ 1.1552,  2.6148,  2.6535,  5.8318,  4.2361])
```

```
>>> torch.normal(2, 3, size=(1, 4))
tensor([[ -1.3987, -1.9544,  3.6048,  0.7909]])
```

Notes & Warnings

NOTE

Note When the shapes do not match, the shape of mean is used as the shape for the returned output tensor

NOTE

Note When std is a CUDA tensor, this function synchronizes its device with the CPU.

Detailed Documentation

Documentation

Returns a tensor of random numbers drawn from separate normal distributions whose mean and standard deviation are given.

The mean is a tensor with the mean of each output element's normal distribution

The std is a tensor with the standard deviation of each output element's normal distribution

The shapes of mean and std don't need to match, but the total number of elements in each tensor need to be the same.

When the shapes do not match, the shape of mean is used as the shape for the returned output tensor

When `std` is a CUDA tensor, this function synchronizes its device with the CPU.

`mean (Tensor)` – the tensor of per-element means

`std (Tensor)` – the tensor of per-element standard deviations

`generator (torch.Generator, optional)` – a pseudorandom number generator for sampling

`out (Tensor, optional)` – the output tensor.

Similar to the function above, but the means are shared among all drawn elements.

`mean (float, optional)` – the mean for all distributions

`std (Tensor)` – the tensor of per-element standard deviations

`out (Tensor, optional)` – the output tensor.

Similar to the function above, but the standard deviations are shared among all drawn elements.

`mean (Tensor)` – the tensor of per-element means

`std (float, optional)` – the standard deviation for all distributions

`out (Tensor, optional)` – the output tensor

Similar to the function above, but the means and standard deviations are shared among all drawn elements. The resulting tensor has size given by `size`.

`mean (float)` – the mean for all distributions

`std (float)` – the standard deviation for all distributions

size (int...) – a sequence of integers defining the shape of the output tensor.

out (Tensor, optional) – the output tensor.